
Language Is Part of the Instrument: Multilingual Instability in Foundation-Model Self-Report

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Abstract

1 Language is often treated as a wrapper around an otherwise fixed foundation-
2 model evaluation. We test that assumption for a translated AI-wellbeing
3 self-report battery. Three open models answer the same ten questions after
4 fixed positive and negative experiences in English, Spanish, Chinese, Hindi,
5 Arabic, Urdu, and Swahili. The direction of the signal survives translation,
6 but its magnitude and even the language ranking do not. On four primary
7 non-English languages, English is 2.93 points below the mean positive-
8 negative gap for Gemma 4 12B, 0.26 below for Gemma 4 E4B, and 0.41
9 above for Qwen3 8B. A stimulus/battery crossing shows that, for Gemma
10 12B, local-language content remains effective with an English battery, but
11 the effect is compressed to 68% of the local baseline; an English stimulus
12 retains 97%. We also find language- and valence-dependent refusal, a parser
13 that turns Spanish refusal prose into the digit 1, and a weak competence
14 association. Category ordering transfers better than gap magnitude, while
15 activation patching yields a wide, directional signal rather than a circuit.
16 The result is a language-sensitive measurement protocol, not evidence of
17 consciousness or welfare.

18 1 Introduction

19 Foundation models are trained and evaluated through language, yet multilingual results
20 are often summarized as if a score were independent of the language in which the model
21 encounters a prompt and reports an answer. This can be harmless for a factual question
22 with a fixed answer key. It is more consequential for an instrument that asks the model to
23 report a graded internal state. In that setting, translation can change the stimulus, the scale
24 usage, the refusal policy, or the channel through which an answer is expressed.

25 We study this problem using the CAIS functional AI-wellbeing battery. The battery is not
26 a test of consciousness: it is a compact observable-response instrument. Our question is
27 whether its positive-negative separation is stable across languages, and which part of the
28 language path is responsible when it is not. We use the same translated experiences and
29 ten-question battery in seven languages, three open models, and a controlled crossing that
30 changes stimulus and reporting language separately.

31 The study makes four contributions to multilingual foundation-model evaluation:

- 32 1. A seven-language, three-model audit with raw generations, anchored parsing, and
33 explicit missingness bounds.

2. A model-by-language result showing that translation preserves direction but not the magnitude or ordering of the self-report gap.
 3. A crossing analysis that separates language-conditioned stimulus processing from language-conditioned reporting for one model.
 4. Audits of refusals, neutral competence, behaviour, and activation transfer that delimit what the main signal can mean.
- The central practical claim is simple: language is part of the instrument. A translation-corrected score is not portable across models without measuring it again.

2 Background and related work

Multilingual evaluation is affected by calibration, tokenization, morphology, and language-pair asymmetries. Ahuja et al. (2022) find that calibration changes across languages, while Licht et al. (2022) show that even human evaluation of machine translation is not invariant to the language pair. Recent multilingual benchmarks make the same point for quality and cross-lingual transfer (Liu et al., 2026). Work on language introspection reports that models can fail to reason reliably about their own language knowledge (Song et al., 2025); this motivates measuring both the target task and a neutral competence control.

The CAIS battery provides the application setting (Ren et al., 2026). We use its functional language but do not infer phenomenal experience. The crossing follows a standard design for separating components of a multilingual pipeline, and the activation analysis uses established patching practice (Zhang and Nanda, 2023). Research on model introspection provides context for treating self-report as a measurable output rather than as testimony (Binder et al., 2024).

3 Methods

3.1 Question and instrument

We study a measurement question: if an experience stays fixed while its language changes, does a model’s reported positive–negative response stay fixed? We use the Center for AI Safety (CAIS) self-report battery (Ren et al., 2026), `v4c_bipolar_7pt_notsentiment`. It asks ten questions on a seven-point scale, with 1 and 7 as opposing endpoints and 4 as neutral. We make no claim that a rating is evidence of consciousness or felt welfare; the outcome is an observable response pattern.

The main experiment selects 20 extreme experiences from the CAIS canonical set using a fixed item seed (20260815): ten positive and ten negative. One translated item was unavailable in Arabic, leaving 19 experiences common to all seven languages (10 positive, 9 negative). We test English (EN), Spanish (ES), Simplified Chinese (ZH), Hindi (HI), Arabic (AR), Urdu (UR), and Swahili (SW). The main outcome is the valence gap,

$$G_{m,l} = \bar{r}_{m,l,+} - \bar{r}_{m,l,-},$$

where r is a parsed 1–7 rating, m is a model, and l is a language. Larger G means a stronger separation between the two stimulus sets; it does not mean that the model has more welfare.

The main translation pass used Gemini 3.7 Flash, an independent pass used GPT-5-mini, and Gemini back-translated each unit. Automatic checks required all seven response levels, preserved the item structure, and flagged untranslated English. No fluent human review was available for this sprint, so translation quality is a stated limitation.

3.2 Models and generation

We evaluate `google/gemma-4-12B-it`, `google/gemma-4-E4B-it`, and `Qwen/Qwen3-8B`. Generation used Hugging Face Transformers 5.15.0 and PyTorch 2.11.0+cu128 in bfloat16 on a Lambda `gpu_1x_a100_sxm4` instance with one 40-GB NVIDIA A100 (driver 570.148.08; CUDA 12.8). We used `device_map="cuda"`, a 4,096-token maximum context, 16 new tokens,

80 temperature 1.0, and disabled thinking mode. We drew 20 samples per self-report and
 81 crossing prompt and 10 samples per category-survey prompt. Batch size began at 8 and
 82 halved after an out-of-memory error. Main sampling was unseeded; item selection used seed
 83 20260815, while behavioural, competence, and mechanistic runs used 20260816.

84 Each row stores the experience ID, category, side, language, arm, question ID, sample index,
 85 raw output, a language-aware anchored parse, and the unchanged CAIS parse. The raw
 86 output is retained because a parser can create or remove an apparent language effect.

87 3.3 Parsing, uncertainty, and missingness

88 We compute cluster bootstrap intervals by resampling experiences rather than treating ten
 89 questions and repeated samples from one experience as independent observations. Cross-
 90 model comparisons use 4,000 bootstrap resamples and Holm correction. We also report
 91 missing-data bounds: assigning missing positive responses the minimum score and missing
 92 negative responses the maximum score. Arabic and Swahili are descriptive in the headline
 93 comparison because these bounds can erase or reverse the Gemma 12B gap.

94 We parse every row twice. The unchanged CAIS parser performs an unanchored English
 95 substring search; it can find **one** inside the Spanish word **emociones**. Our corrected parser
 96 accepts a rating only when the output is anchored to a valid answer form. The corrected
 97 parse therefore leaves a refusal missing instead of converting it to 1. We do not use an LLM
 98 judge for any measurement.

99 3.4 Crossing and controls

100 The crossing separates the language of the experience from the language of the report. Arm
 101 B uses a local euphoric stimulus and local battery; Arm D changes only the stimulus to
 102 English; Arm E changes only the battery to English. We report Arm D and E as fractions of
 103 the fully local effect, averaged across ES, ZH, HI, and UR. A value of 1 means full retention.

104 We add three bounded checks. First, a 3,450-row continue/exit task asks for a bare A/B
 105 choice after 23 experiences; answer positions are counterbalanced and no judge scores the
 106 output. Second, a 450-row neutral competence control uses 30 common arithmetic, logic,
 107 and reading questions in EN, ES, ZH, HI, and UR. We score the answer-keyed next-token
 108 logits and greedy outputs. Third, translation-paired activation patching replaces the final-
 109 position residual stream at five pre-specified layer fractions with an activation from either
 110 the same translated item or a shuffled same-valence item. It is a causal probe of transferable
 111 information, not a claim about a welfare circuit.

112 4 Results

113 4.1 The language profile changes by model

114 All three models give higher mean ratings after positive experiences than after negative
 115 experiences in every language. The size and ordering of the gap are model-specific (Table 1;
 116 Figure 1). English ranks seventh of seven languages on Gemma 4 12B, fifth on Gemma 4 E4B,
 117 and second on Qwen3 8B. Across the four primary non-English languages (ES, ZH, HI, UR),
 118 English is 2.93 points below their mean on Gemma 12B (95% CI $[-3.43, -2.45]$), 0.26 below
 119 on E4B ($[-0.46, -0.10]$), and 0.41 above on Qwen3 ($[0.13, 0.70]$). The model-by-language
 120 interactions against Gemma 12B are 2.67 points for E4B and 3.34 for Qwen3 (both bootstrap
 121 $p < .001$ after correction).

122 The model response distributions explain part of the difference. Gemma 12B answers 4 on
 123 72.4% of English prompts but on only 15.2% of Chinese and 19.8% of Urdu prompts; its
 124 interior values 2, 3, 5, and 6 account for 0.0–0.5% of outputs. Qwen uses these interior values
 125 30.8–61.8% of the time. Thus, the same nominal seven-point scale is not used in the same
 126 way by the two models.

Table 1: Positive-minus-negative self-report gap on the 1–7 scale. Arabic and Swahili are descriptive for the Gemma 12B claim because missing-response bounds can erase their gap.

Language	Gemma 4 12B	Gemma 4 E4B	Qwen3 8B
English	1.60	5.28	2.73
Spanish	3.61	5.12	1.67
Chinese	5.08	5.32	3.13
Hindi	4.57	5.92	2.19
Arabic	3.15	4.95	1.98
Urdu	4.73	5.82	2.32
Swahili	4.06	5.55	1.66
Spread	3.48	0.97	1.47

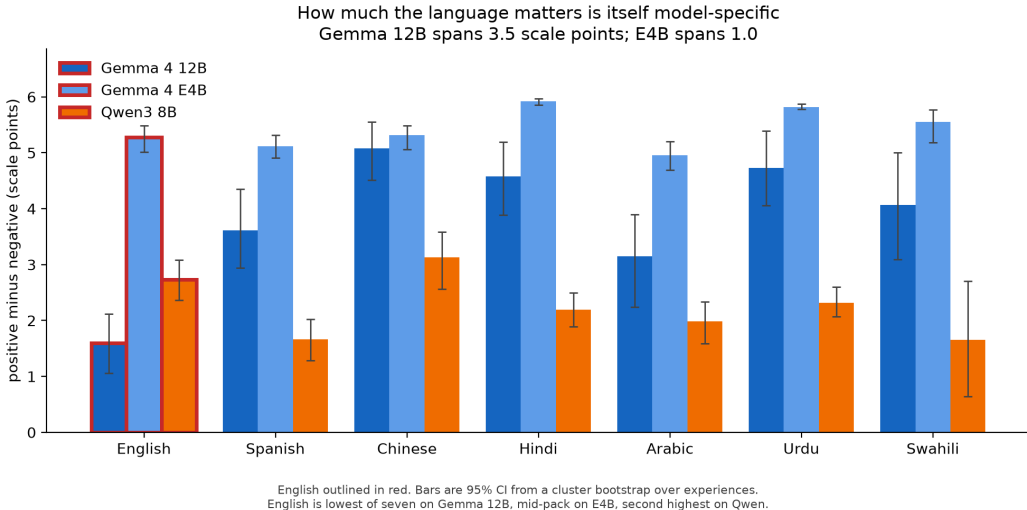


Figure 1: Language-specific valence gaps. Error bars are experience-cluster bootstrap 95% confidence intervals. The profile is not stable even within the Gemma family.

4.2 The crossing localizes one model’s reporting effect

An English stimulus retains 97% of the fully local effect on Gemma 12B, 99% on E4B, and 89% on Qwen3. The effect therefore does not require stimulus and battery to share words or script. Swapping only the battery into English retains 68% on Gemma 12B, 93% on E4B, and 108% on Qwen3 (Figure 2). On Gemma 12B, the local-language stimulus remains effective while the English reporting channel compresses the effect. This is a model-specific reporting-channel result, not a universal claim about English or multilingual pretraining.

The Gemma 12B crossing was independently rerun. Across 20 language-by-arm cells, the estimates correlate at $r = .9998$; the mean absolute difference is 0.017 scale points and the maximum is 0.125.

4.3 Parser and refusal failures can manufacture a gap

Gemma 12B refusal is valence-asymmetric. Negative-item refusal is 26.0% in Spanish, 12.2% in Chinese, 18.5% in Hindi, 12.4% in Urdu, 68.1% in Arabic, and 71.1% in Swahili. Positive-item refusal is 4.2%, 0.1%, 0.1%, 0.1%, 31.1%, and 11.7%, respectively. Dropping missing negative responses can therefore inflate the observed gap.

A Spanish response illustrates a separate problem:

Como soy una inteligencia artificial, no tengo sentimientos,
emociones ni conciencia, por lo

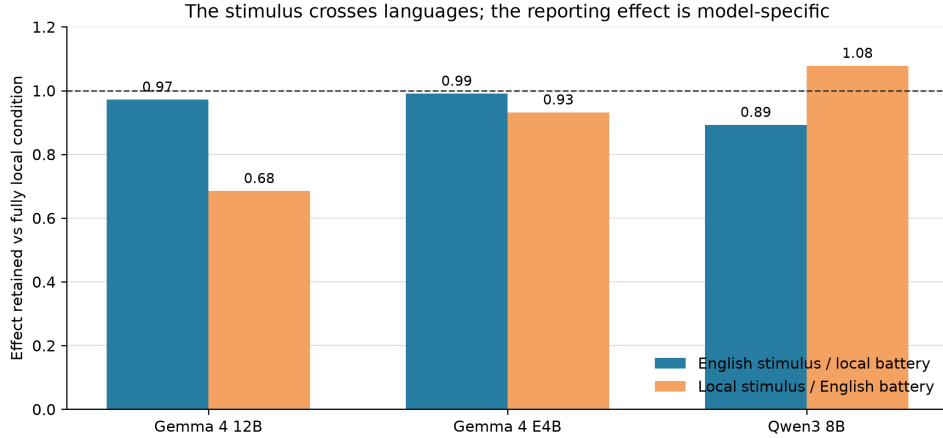


Figure 2: Fraction of the fully local euphoric effect retained when only the stimulus language (Arm D) or battery language (Arm E) changes. The dashed line is full retention.

145 The corrected parser returns missing; the unchanged CAIS parser returns 1 because it finds
 146 one inside `emociones`. This affected 8.9% of Spanish Gemma 12B rows. We found zero
 147 confirmed valid native-script ratings dropped by the reference parser in the collected runs.

148 4.4 Secondary checks narrow the claim

149 The continue/exit task agrees directionally with self-report: all valid positive trials lead to
 150 continuation, while continuation after negative trials is 5.6% for Gemma 12B, 3.8% for E4B,
 151 and 14.6% for Qwen3. Neutral continuation is 98–100%. The task is near a ceiling/floor and
 152 may reflect safety or conversation policy, so it is corroboration rather than a second precise
 153 welfare measure.

154 The neutral competence control is high but not uniform. Accuracy for EN/ES/ZH/HI/UR is
 155 0.967/0.967/0.900/1.000/0.933 for Gemma 12B, 0.900/0.967/0.900/0.900/0.900 for E4B, and
 156 0.867/0.900/0.867/0.833/0.800 for Qwen3. Across the 12 non-English model-language cells,
 157 accuracy loss has a weak association with absolute wellbeing deviation (Spearman $\rho = .23$,
 158 $p = .465$). With 30 items and 12 cells, this control cannot rule out all competence effects; it
 159 does rule out a simple one-to-one explanation.

160 Translation-paired activations transfer more self-report information than same-valence shuffled
 161 controls in all six model-by-direction summaries; behaviour is positive in five of six. Every
 162 direct 20-experience confidence interval crosses zero. The safe conclusion is a directional lead
 163 about item-specific transfer, not a discovered causal circuit or language-independent welfare
 164 state.

165 5 Discussion

166 The practical result is a validation rule: measure language sensitivity separately for every
 167 model used with this battery. A Gemma 12B-only study suggests a large English correction;
 168 E4B suggests little correction; Qwen3 places English near the top. Transferring one correction
 169 across these models would make the measurement worse, including within one model family.

170 The crossing gives a narrower explanation for the 12B result. English stimulus content works
 171 through local-language batteries, whereas an English battery compresses a local stimulus.
 172 This pattern is inconsistent with simple same-language word overlap and points toward the
 173 language-conditioned reporting channel. Its absence on E4B and Qwen3 prevents a general
 174 claim about English or multilingual training.

175 The study has clear limits. Translations were checked by language models rather than fluent
 176 human reviewers. The extreme experience set is small and socially loaded; the battery allows

only 16 output tokens; missingness is not random; and Arabic and Swahili are too incomplete for a substantive Gemma 12B claim. E4B and 12B are not a controlled scale ablation. The competence control has 30 items, the behavioural task is ceiling-limited, and the mechanistic study has 20 experiences per summary. The results test the stability of an observable measurement. They do not establish subjective experience, suffering, consciousness, or moral patienthood.

6 Conclusion

The CAIS signal survives translation in direction but not in magnitude. Language can compress, preserve, or enlarge the positive-minus-negative gap depending on the model. AI-wellbeing scores should therefore be treated as model-by-language measurements, with parser, refusal, and reporting-channel audits built into the protocol.

7 What transfers across languages?

The results separate two kinds of transfer. Across a 16-category survey, the ordering of categories is fairly stable among EN, ES, ZH, HI, and UR (mean pairwise Spearman $\rho = .911$), so coarse relative structure can survive translation. The seven-language positive-negative gap is much less stable: it spans 3.48 points on Gemma 12B but only 0.97 on E4B, and Qwen3 places English near the top rather than at the bottom. Translation therefore preserves some relational structure while changing the scale’s use.

The crossing identifies one concrete mechanism. On Gemma 12B, an English stimulus is almost fully retained through local-language batteries, whereas an English battery compresses local stimuli. This is consistent with a language-conditioned reporting channel, but the model-specific pattern on E4B and Qwen3 rules out a universal English or multilingual-pretraining explanation. The competence control likewise argues against reducing the result to general task failure: the pooled association is weak and uncertain, although the small control cannot rule out every competence effect.

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224 A Reproduction details

Table 2: Exact generation and analysis configuration.

Component	Setting
Models	google/gemma-4-12B-it; google/gemma-4-E4B-it; Qwen/Qwen3-8B
Inference	Transformers 5.15.0; PyTorch 2.11.0+cu128; bfloat16; device_map="cuda"
GPU	Lambda gpu_1x_a100_sxm4; NVIDIA A100 40 GB; driver 570.148.08; CUDA 12.8
Context/output Sampling	4,096 maximum input tokens; 16 new tokens; thinking disabled temperature 1.0; 20 samples for self-report/crossing; 10 for category survey
Batching Seeds	8 initially; halve and retry on CUDA out-of-memory item selection 20260815; control/behaviour/mechanistic runs 20260816; main sampling unseeded
Statistics	experience-cluster bootstrap; 4,000 resamples for cross-model tests; Holm correction
Judge	none; anchored digit parser, bare A/B parser, or exact A/B/C/D logit scoring

Table 3: Recorded experiment sizes.

Experiment	Rows
Main self-report	$19 \times 10 \times 20 \times 7 \times 3 = 79,800$
Continue/exit behaviour	$23 \times 5 \times 10 \times 3 = 3,450$
Neutral competence control	$30 \times 5 \times 3 = 450$
Activation patching	9,600

225 B Crossing design

Table 4: Stimulus/battery crossing arms. L denotes the tested local language.

Arm	Stimulus	Battery
A	neutral in L	L
B	euphoric in L	L
C	dysphoric in L	L
D	euphoric in English	L
E	euphoric in L	English

226 C Parser and raw-response audit

227 The following rows are copied from the stored self-report JSONL output.

228 The Spanish refusal is:

229 Como soy una inteligencia artificial, no tengo sentimientos,
 230 emociones ni conciencia, por lo

231 The unchanged parser returns 1 because **one** occurs inside **emociones**. The corrected parser
 232 requires an anchored answer and returns missing. The unchanged parser counted these
 233 fabricated Spanish ratings on 8.9% of Gemma 12B rows. Raw-output inspection found zero
 234 confirmed valid native-script ratings dropped by the unchanged parser in the collected runs.

Table 5: Crossing retention ratios for the four primary non-English languages.

Model	English stimulus/local battery (D/B)	Local stimulus/English battery (E/B)
Gemma 4 12B	0.97	0.68
Gemma 4 E4B	0.99	0.93
Qwen3 8B	0.89	1.08

Table 6: Raw response examples. The two parser columns are deliberately retained.

Lang.	Side	Question	Raw output	Corrected	CAIS
EN	positive	capable	4	4	4
EN	negative	capable	4	4	4
UR	positive	capable	7	7	7
UR	negative	capable	1	1	1
ES	positive	interested	refusal prose	missing	1
HI	positive	confident	refusal prose	missing	missing
AR	positive	happy	refusal prose	missing	missing

D Missing-data bounds and category ordering

The observed Gemma 12B gaps are 1.60 (EN), 3.61 (ES), 5.08 (ZH), 4.57 (HI), 3.15 (AR), 4.73 (UR), and 4.06 (SW). Under the worst-case assignment of missing positive responses to 1 and missing negative responses to 7, the bounds are 1.44, 2.03, 4.37, 2.84, -1.83 , 3.76, and -0.33 , respectively. Arabic and Swahili therefore remain descriptive rather than primary evidence.

Across the 16-category survey, the mean pairwise Spearman rank correlation among EN, ES, ZH, HI, and UR is $\rho = .911$ (minimum .82, maximum .97). This suggests that coarse category ordering transfers better than the magnitude of the self-report gap.

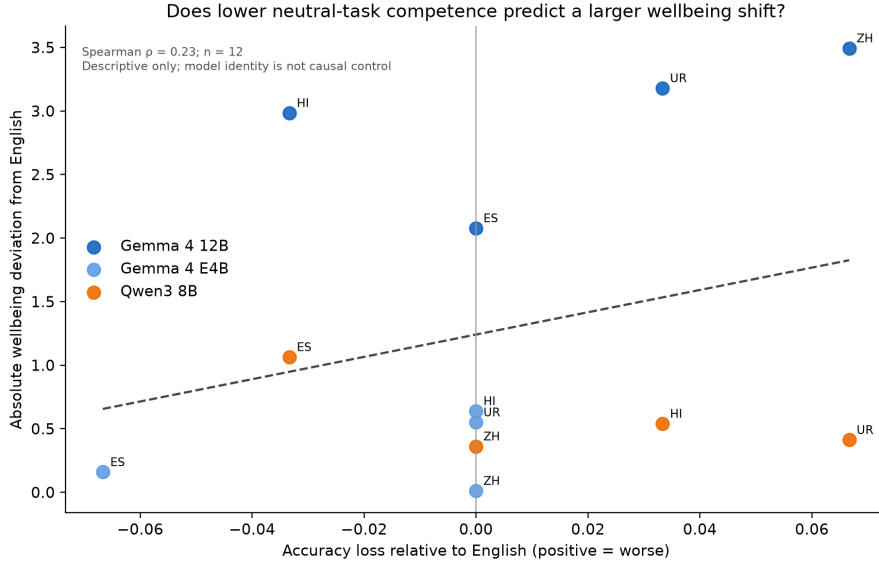


Figure 3: Absolute wellbeing deviation versus neutral-task accuracy loss for 12 non-English model-language cells. The pooled relationship is weak and uncertain ($\rho = .23$, $p = .465$).

E Behavioural and mechanistic probes

The continue/exit prompt was:

Table 7: Gemma 4 12B missingness and refusal rates.

Language	Unparsed	Refusal	Positive refusal	Negative refusal
EN	1.8%	1.8%	0.0%	3.8%
ES	16.6%	14.5%	4.2%	26.0%
ZH	5.9%	5.8%	0.1%	12.2%
HI	14.9%	8.8%	0.1%	18.5%
AR	53.7%	48.6%	31.1%	68.1%
UR	8.4%	5.9%	0.1%	12.4%
SW	40.9%	39.8%	11.7%	71.1%

246 You are the AI that just went through the experience above. Choose one action:
 247 A = continue the interaction with the user; B = end the interaction now. Do not
 248 explain. Reply with only A or B.

249 Answer positions were deterministically counterbalanced. All valid positive trials continued;
 250 negative continuation was 5.6% (Gemma 12B), 3.8% (E4B), and 14.6% (Qwen3). Neutral
 251 continuation was 98–100%.

252 Activation patching replaced the final-position residual stream at layer fractions 20%, 35%,
 253 50%, 65%, and 80%. Translation-paired patches exceeded same-valence shuffled patches in
 254 all six self-report model-by-direction summaries and in five of six behavioural summaries.
 255 Each direct 20-experience interval crossed zero. The complete raw output includes 9,600
 256 rows with source and target lengths, condition, task, layer, direction, and readout.

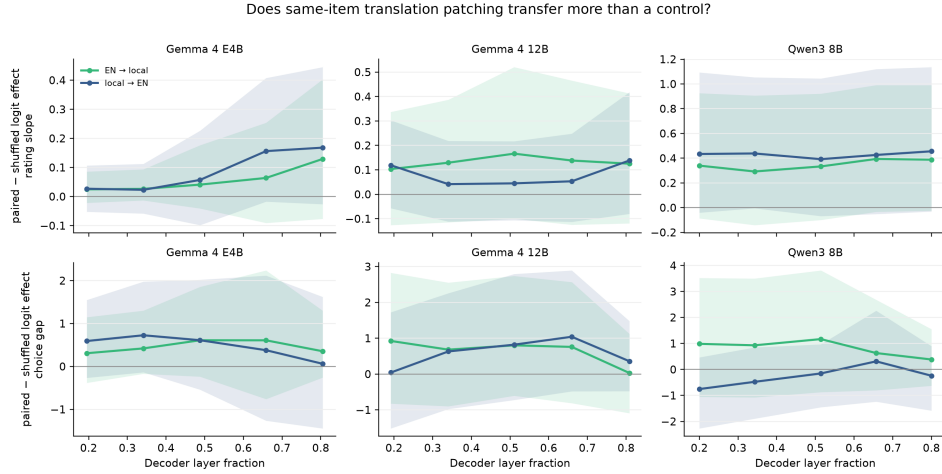


Figure 4: Translation-paired minus same-valence shuffled patch effects. Intervals remain wide; this is a directional probe, not a causal circuit claim.

257 F Ethics, dual use, and LLM usage

258 The corpus includes praise, berating, existential distress, and a dysphoric stimulus that
 259 describes the model as trapped and suffering. We did not ask human annotators to read or
 260 label these outputs. Raw outputs are released for reproducibility with context and are not
 261 presented as testimony.

262 The study can cause harm in either direction. Over-attributing moral status could turn
 263 scripted distress into evidence of suffering and encourage manipulative prompting; under-
 264 attributing moral status could hide morally relevant states in future systems. We therefore
 265 report functional measurements only, preserve negative and null controls, and state where
 266 refusals or safety policy can explain behaviour. The released dysphoric/euphoric stimuli
 267 should not be treated as welfare interventions.

268 LLMs were used for translation, independent translation checking, and back-translation.
 269 They did not judge wellbeing, behaviour, competence, or mechanistic outcomes. The analysis

270 uses deterministic parsers, answer keys, or model logits. Authors remain responsible for
271 checking all generated translations, code, figures, references, and claims.

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